Advanced Causal Inference

Lecture 10: Semiparametric Efficiency Theory (Semiparametric Model)

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This lecture note is partially based on the lecture notes/textbook of Andrea Rotnitzky

Outline

1. Review
2. Extension to Semiparametric Models
3. Examples

Review

- In a regular parametric model, the variance of any unbiased estimator attains the CR bound
- The CR bound does not necessarily apply to consistent estimators,
  as superefficient estimators can outperform the CR bound at specific parameter values
- For regular estimators, the CR bound holds
- Asymptotically linear estimators admit a linear representation with an influence function (IF),
  but not all AL estimators are regular
- If an IF satisfies (IEF), the corresponding estimator is regular and asymptotically linear (RAL)

$$\left.\frac{\partial \beta(\theta)}{\partial \theta^{\intercal}}\right|_{\theta=\theta^*} = \mathrm{E}_{\theta^*}\left\{\varphi_{\theta^*}(Z)\boldsymbol{S}(Z,\theta^*)^{\intercal}\right\} \qquad \text{(IFE)}$$

- Projecting the IF of a RAL estimator onto the tangent space yields the efficient influence function (EIF)
- The RAL estimator corresponding to the EIF attains the CR bound

Goal

- So far, our discussion has been restricted to parametric models
- Today, we will turn to semiparametric models, which include nonparametric models as a special case

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2 Extension to Semiparametric Models

3 Examples

Regular Parametric Submodel

- Given a semiparametric model $\mathcal{F}$, we say that $\mathcal{F}_{\text{sub}}$ is a regular parametric submodel at F if and only if
  - $F \in \mathcal{F}_{\text{sub}}$
  - $\mathcal{F}_{\text{sub}} \subseteq \mathcal{F}$
  - $\mathcal{F}_{\text{sub}}$ is regular

- Example: $\mathcal{F} = \{\text{linear model } Y = X^\intercal \beta + \epsilon \text{ with } \mathrm{E}(\epsilon|X) = 0\}$ with $F \in \mathcal{F}$
  - $\mathcal{F}_{\text{sub}} = \{Y = X^\intercal \beta + \epsilon, \epsilon \sim N(0, \sigma^2)\}$
  - $\mathcal{F}'_{\text{sub}} = \{Y = X^\intercal \beta + \epsilon, \epsilon \sim N(0, \sigma^2(X))\}$
  - $\mathcal{F}''_{\text{sub}} = \{Y = X^\intercal \beta + \epsilon, \epsilon \sim t(5)\}$
  - $\mathcal{F}'''_{\text{sub}} = \{Y = X^\intercal \beta + X^{\otimes 2\intercal} \gamma + \epsilon, \sim N(0, \sigma^2)\}$: $\mathcal{F}'''_{\text{sub}} \nsubseteq \mathcal{F}$

- From now on, all parametric submodels we refer to are regular parametric submodels

Tangent Space

- Let $\mathcal{A}$ be a collection of reg. parametric submodels of a semiparametric model $\mathcal{F}$ through F:

$$\mathcal{A} = \left\{ \mathcal{F}_{\text{sub}} \,\middle|\, \mathcal{F}_{\text{sub}} : \text{reg. para. submodel of } \mathcal{F} \text{ through } F \right\}$$

- The tangent space of model $\mathcal{F}$ w.r.t. $\mathcal{A}$ at F is defined as

$$\Lambda_{\mathcal{F}}(F) = \text{closure}\left[\bigcup_{\mathcal{F}_{\text{sub}} \in \mathcal{A}} \Lambda_{\mathcal{F}_{\text{sub}}}(F) \right]$$

- As a general rule, to construct the tangent space, we start with an "educated guess" guided by the restrictions that the scores must satisfy.

- These restrictions are deduced from the constraints of the semiparametric model.

- Verifying that the conjectured space is indeed the tangent space is a technical exercise, which consists of:

(a) showing that the scores of regular submodels lie in the postulated set (easy)

(b) exhibiting, for any given element of the postulated set, a sequence of regular parametric submodels such that linear combinations of the scores in the sequence converge to the given element (tedious)

Example: Tangent Space of the Nonparametric Model

- Let $\mathcal{F}_{\text{NP}}$ be the nonparametric model, which does not put any restriction on the observed data
- Let $\mathcal{A}$ be a collection of all reg. parametric submodels of $\mathcal{F}_{\text{NP}}$
- The set of scores of all regular parametric submodels should satisfy no restrictions other than having mean zero and finite variance
- **Educated guess**: the tangent space at a given law F is equal to

$$\mathcal{L}_2^0(F) = \left\{ S(Z) \,|\, \mathrm{E}_F(Z) = 0 \ , \ \mathrm{Var}_F(Z) < \infty \right\}$$

- This is indeed true

Example: Semiparametric Model with Independence

- Let $\mathcal{F}$ be a collection of distributions on (X, Y) with $X \perp\!\!\!\perp Y$, i.e., the density satisfies $f(X,Y) = f_X(X) f_Y(Y)$
- A possible choice of a parametric submodel for $\mathcal{F}$ would be

$$\mathcal{F}_{\text{sub}} = \left\{ f(X,Y,\eta_1,\eta_2) \,\middle|\, f(X,Y,\eta_1,\eta_2) = f_X(X,\eta_1) f_Y(Y,\eta_2) \right\}$$

  where η_1 and η_2 parametrize the distributions of X and Y, respectively

- Then, the tangent space under this submodel is

$$\Lambda_{\mathcal{F}_{\text{sub}}} = \left\{ a_1 S_X(X,\eta_1) + a_2 S_Y(Y,\eta_2) \,\middle|\, a_1, a_2 \in \mathbb{R} \right\}$$

  where $S_X(X,\eta_1)$ and $S_Y(Y,\eta_2)$ are score functions of the densities of X and Y in $\mathcal{F}_{\text{sub}}$, respectively

- **Educated guess**: Since one can choose different parametric models for (f_X, f_Y), their gradients $(=S_X, S_Y)$ are unrestricted except the restrictions on the score function

$$\Lambda_{\mathcal{F}}(F) = \text{closure}\left[\bigcup_{\mathcal{F}_{\text{sub}} \in \mathcal{A}} \Lambda_{\mathcal{F}_{\text{sub}}}(F) \right] = \left\{ S_X(X) + S_Y(Y) \,\middle|\, \begin{array}{l} \mathrm{E}_F\{S_X(X)\} = \mathrm{E}_F\{S_Y(Y)\} = 0 \\ \mathrm{Var}_F\{S_X(X)\} < \infty, \mathrm{Var}_F\{S_Y(Y)\} < \infty \end{array} \right\}$$

Role of $\mathcal{A}$

- The efficiency theory we will discuss is always relative to the choice of $\mathcal{A}$
- In many cases, one can take $\mathcal{A}$ to be a "maximal class":

$$\mathcal{A}_{\text{max}} = \{\text{all regular parametric submodels of } \mathcal{F} \text{ passing through } F\}$$

- Although $\mathcal{A}_{\text{max}}$ may not always be technically valid, we will ignore this issue and proceed by considering the maximal class and omit explicit reference to $\mathcal{A}$ in what follows
- For simplicity, we let $\mathcal{F}_{\text{sub}}$ denote a generic regular parametric submodel

Regular Estimator

Definition (Regular Estimator)

An estimator sequence $\widehat{\beta}_N$ is said to be a regular estimator of $\beta(F) \in \mathbb{R}^k$ in model $\mathcal{F}$ at F if it is a regular estimator at F under every $\mathcal{F}_{\text{sub}}$

- From the Hajek's convolution thm, a regular estimator within a reg. parametric model has a variance bound:

$$\mathrm{Var}_F\left[\sqrt{N}\{\widehat{\beta}_N - \beta(F)\}\right] \geqslant C_{\mathcal{F}_{\text{sub}}}(F)$$

- Therefore,

$$\sup_{\mathcal{F}_{\text{sub}}} C_{\mathcal{F}_{\text{sub}}}(F) \leqslant \mathrm{Var}_F\left[\sqrt{N}\{\widehat{\beta}_N - \beta(F)\}\right]$$

- Even though the LHS is finite, it does not necessarily guarantee the existence of a regular estimator

  But if the LHS is infinite, a regular estimator does not exist

$$\text{A regular estimator exists} \quad \Rightarrow \quad \sup_{\mathcal{F}_{\text{sub}}} C_{\mathcal{F}_{\text{sub}}}(F) < \infty$$

- When does $\sup_{\mathcal{F}_{\text{sub}}} C_{\mathcal{F}_{\text{sub}}}(F) < \infty$ hold?

Pathwise Differentiability

Definition (Pathwise Differentiability)

Let $\mathcal{F}$ be a semiparametric model and $F^* \in \mathcal{F}$ be a law

We call $\beta : \mathcal{F} \to \mathbb{R}^k$ a pathwise differentiable (or regular) parameter at F^* iff there exists a function $\varphi_{F^*}(Z) \in \mathcal{L}_2(F^*)$ satisfying the following condition:

$\forall \mathcal{F}_{\text{sub}} = \{F_\theta\}$ with $F^* = F_{\theta^*}$ and score $\boldsymbol{S}(Z, \theta^*)$, we have

$$\left.\frac{\partial \beta(F_\theta)}{\partial \theta^\intercal}\right|_{\theta=\theta^*} = \mathrm{E}_{F^*}\left[\varphi_{F^*}(Z)\boldsymbol{S}(Z,\theta^*)^\intercal\right] \qquad \text{(IFE-Semi)}$$

- (IFE-Semi) is an extension of the (IFE-Para) to semiparametric models

Definition (Gradient / Influence Function)

The function $\varphi_F(Z)$ in (IFE-Semi) is referred to as the gradient of $\beta(\cdot)$ at F

If $\mathrm{E}_F\{\varphi_F(Z)\} = 0$, we also call it an influence function (IF)

Pathwise Differentiability

- Suppose that $\beta : \mathcal{F} \to \mathbb{R}$ is pathwise differentiable

- Then, it admits the **von Mises representation** which a kind of distributional Taylor expansion

$$\begin{aligned} \beta(G) - \beta(F) &= \int \varphi_G(z)\, d(G-F)(z) + R_2(G,F) \\ \beta(a) - \beta(b) &= \beta'(a)(a-b) + O(\|a-b\|^2) \end{aligned}$$

- If we take $G = F + \epsilon F'$ and take $\epsilon \to 0$, we have

$$\left.\frac{\partial \beta(F+\epsilon F')}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{E}_F\big[\varphi_F(Z) S(Z,F)\big] + \frac{\partial}{\partial \epsilon} R_2(F+\epsilon F', F)\ , \quad \forall F'$$

- From the definition of the IF, we have

$$\frac{\partial}{\partial \epsilon} R_2(F+\epsilon F', F) = 0\ , \quad \forall F'$$

- This implies that R_2 is at most of second order with respect to the change in the distribution

Semiparametric CR Bound

- Within each **reg. para.** submodel, we have

$$C_{\mathcal{F}_{\text{sub}}}(\theta) = \operatorname{Var}_\theta\left[\Pi_\theta\left\{\varphi_F(Z)|\Lambda_{\mathcal{F}_{\text{sub}}}(\theta)\right\}\right]$$

- This implies

$$\begin{aligned}\sup_{\mathcal{F}_{\text{sub}}} C_{\mathcal{F}_{\text{sub}}}(F) &= \sup_{\mathcal{F}_{\text{sub}}} C_{\mathcal{F}_{\text{sub}}}(\theta) \\ &= \sup_{\mathcal{F}_{\text{sub}}} \operatorname{Var}_\theta\left[\Pi_\theta\left\{\varphi_F(Z)|\Lambda_{\mathcal{F}_{\text{sub}}}(\theta)\right\}\right] \\ &\leqslant C_{\mathcal{F}}(F) := \operatorname{Var}_F\left[\Pi_F\left\{\varphi_F(Z)|\Lambda_{\mathcal{F}}(F)\right\}\right] < \infty\end{aligned}$$

- The supremum of the CR bounds over all $\mathcal{F}_{\text{sub}}$ for a pathwise differentiable parameter $\beta(\cdot)$ is finite

- And, it is $\leqslant$ Var[projection of any IFs for the parameter into the tangent space for the model]

- Semiparametric CR bound aka Semiparametric Efficiency Bound

- $\varphi_{F,\text{eff}}(Z) := \Pi_F\left\{\varphi_F(Z)|\Lambda_{\mathcal{F}}(F)\right\}$ = the (unique) efficient influence function (EIF) for $\beta(F)$ in model $\mathcal{F}$ at F

Convolution Theorem

Theorem (Convolution Theorem (van der Vaart, 1998, Theorem 25.20))

Let $\mathcal{F}$ be a semiparametric model and $\beta : \mathcal{F} \to \mathbb{R}^k$ be a path. diff. para at F with the EIF $\varphi_F(Z)$.
If $\widehat{\beta}_N$ is a regular estimator of $\beta(\theta)$ at F, then

$$\sqrt{N}\{\widehat{\beta}_N - \beta(\theta)\} \overset{D(F)}{\to} U + V$$

Here, $U \sim N_k(0, C_{\mathcal{F}}(F))$ is the SEB for $\beta(F)$ in $\mathcal{F}$ and V is independent of U

- Therefore, a regular estimator $\widehat{\beta}_N$ is most efficient when $V = 0$
- The SEB is the smallest variance attainable by any regular estimator in the semiparametric model $\mathcal{F}$

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If $\widehat{\beta}_N$ is a regular estimator of $\beta(\theta)$ at F, then

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Here, $U \sim N_k(0, C_{\mathcal{F}}(F))$ where $C_{\mathcal{F}}(F) = \text{Var}_F\{\varphi_{F,\text{eff}}(Z)\}$ is the SEB for $\beta(F)$ in $\mathcal{F}$ and V is independent of U

- Therefore, a regular estimator $\widehat{\beta}_N$ is most efficient when $V = 0$
- The SEB is the smallest variance attainable by any regular estimator in the semiparametric model $\mathcal{F}$

Asymptotically Efficient Estimator

Definition (Asymptotically Efficient Estimator)

Let $\mathcal{F}$ be a semiparametric model and $\mathcal{F}' \subseteq \mathcal{F}$ and $\beta : \mathcal{F} \to \mathbb{R}^k$ be a path. diff. para at each $F \in \mathcal{F}$

- We call an estimator $\widehat{\beta}_N$ **locally** asymptotically efficient in $\mathcal{F}$ at $\mathcal{F}'$ if
  1. it is regular in $\mathcal{F}$; and
  2. it is asymptotically efficient estimator in $\mathcal{F}$ at $\forall F \in \mathcal{F}'$, i.e.,

$$\sqrt{N}\{\widehat{\beta}_N - \beta(F)\} \overset{D(F)}{\to} N_k(0, C_{\mathcal{F}}(F)) \ , \quad \forall F \in \mathcal{F}'$$

- We call a regular estimator $\widehat{\beta}_N$ **globally** asymptotically efficient if $\mathcal{F}' = \mathcal{F}$

- Locally (much less globally) efficient estimators of a pathwise differentiable parameter do not always exist

Regular and Asymptotically Linear Estimator

- However, if an efficient estimator exists, it must be RAL with the EIF as an IF

Lemma 25.23 of van der Vaart (1998)

Let $\mathcal{F}$ be a semiparametric model and $\beta : \mathcal{F} \to \mathbb{R}^k$ be a path. diff. para at F with the EIF $\varphi_{F,\text{eff}}(Z)$

The estimator $\widehat{\beta}_N$ is locally asymptotically efficient at F if and only if it is RAL with the following expression:

$$\sqrt{N}\{\widehat{\beta}_N - \beta(F)\} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \varphi_{F,\text{eff}}(Z_i) + o_{P,F}(1)$$

- This result states, just as in the parametric case, that as far as efficiency is concerned, we don not loose anything by restricting attention to RAL estimators

Summary

- We say $\beta(F)$ is pathwise differentiable with an IF $\varphi_F(Z)$ if

$$\left.\frac{\partial \beta(F_\theta)}{\partial \theta^\intercal}\right|_{\theta=\theta^*} = \mathrm{E}_{F^*}\left[\varphi_{F^*}(Z)\boldsymbol{S}(Z,\theta^*)^\intercal\right]$$

  which results in a term is at most of second order with respect to the change in the distribution

- The EIF is the projection of φ_F onto the tangent space of $\mathcal{F}$:

$$\varphi_{F,\text{eff}}(Z) := \Pi_F\left\{\varphi_F(Z)|\Lambda_{\mathcal{F}}(F)\right\}$$

- The variance of the EIF is called the SEB:

$$C_{\mathcal{F}}(F) := \mathrm{Var}_F\left\{\varphi_{F,\text{eff}}(Z)\right\}$$

- The SEB is the variance bound of regular estimators in $\mathcal{F}$

An estimator is locally asymptotically efficient at F

$\Leftrightarrow$ it is RAL with the IF $= \varphi_{F,\text{eff}}$

$\Leftrightarrow$ it attains the SEB

Takeaway

- A principled procedure for constructing an estimator:

1. Specify a causal or statistical model $\mathcal{F}$
2. Identify a functional $\tau(F)$ that characterizes the causal parameter of interest
3. Derive an IF, or if possible, the EIF, for $\tau(F)$ within the model $\mathcal{F}$
4. Construct $\widehat{\tau}$, an estimator for $\tau(F)$ based on the (E)IF
5. Find conditions under which $\widehat{\tau}$ is RAL

- Key properties of the resulting estimator:
  - An IF-based estimator of $\tau(F)$ is robust to bias in the nuisance component estimators i.e., exhibits Neyman orthogonality (Chernozhukov et al., 2018)
  - An EIF-based estimator of $\tau(F)$ achieves the SEB, meaning that
    1. it is the most efficient RAL estimator of $\tau(F)$ in model $\mathcal{F}$; and
    2. further efficiency gains require imposing stronger assumptions on the model

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Example: Population Mean in the Nonparametric Model

1. Specify a causal or statistical model $\mathcal{F}$

- $\mathcal{F} =$ nonparametric model, $\beta(F) = \mathrm{E}_F(Z)$

2. Identify a functional $\tau(F)$ that characterizes the causal parameter of interest

- There is no potential outcome, so $\mathrm{E}_F(Z)$ is an identifying functional

Example: Population Mean in the Nonparametric Model

3. Derive an IF, or if possible, the EIF, for $\tau(F)$ within the model $\mathcal{F}$

- Consider $\mathcal{F}_{\text{sub}} = \{F_\theta\}$ with $F_{\theta *} = F^*$, $\beta(F_\theta) = \mathrm{E}_\theta(Z)$
- Calculate the gradient:

$$\frac{\partial}{\partial \theta^\intercal}\beta(F_\theta) = \frac{\partial}{\partial \theta^\intercal}\mathrm{E}_\theta(Z) = \frac{\partial}{\partial \theta^\intercal}\int z f(z,\theta)\,dz$$
$$= \int z \nabla_\theta f(z,\theta)\,dz = \int z \frac{\nabla_\theta f(z,\theta)}{f(z,\theta)} f(z,\theta)\,dz = \int z \boldsymbol{S}(z,\theta) f(z,\theta)\,dz = \mathrm{E}_\theta[Z\boldsymbol{S}(Z,\theta)]$$

- Since $\mathrm{E}_\theta[\boldsymbol{S}(Z,\theta)] = 0$, we have

$$\frac{\partial}{\partial \theta^\intercal}\beta(F_\theta) = \mathrm{E}_\theta[\{Z - \mathrm{E}_\theta(Z)\}\boldsymbol{S}(Z,\theta)] \quad \Rightarrow \quad \frac{\partial}{\partial \theta^\intercal}\beta(F_\theta)\Big|_{\theta=\theta^*} = \mathrm{E}_{F^*}[\{Z - \mathrm{E}_{F^*}(Z)\}\boldsymbol{S}(Z,\theta^*)]$$

  implying that $\varphi_F(Z) = Z - \beta(F)$ is an IF of $\beta(F) = \mathrm{E}_F(Z)$

- Recall that $\Lambda_{\mathcal{F}}(F) = \mathcal{L}_2^0(F)$, i.e., all mean zero and sq. integrable fts. Therefore, $\varphi_F(Z) \in \mathcal{L}_2^0(F)$, implying that

$$\varphi_{F,\text{eff}}(Z) = \Pi_F\big[\varphi_F(Z) \,\big|\, \Lambda_{\mathcal{F}}(F)\big] = \varphi_F(Z) = Z - \beta(F)$$

Example: Population Mean in the Nonparametric Model

4. Construct $\widehat{\tau}$, an estimator for $\tau(F)$ based on the (E)IF

- Since $\varphi_{F,\text{eff}}(Z) = Z - \beta(F)$, we find an estimator of $\beta(F)$ by solving an moment equation:

$$0 = \frac{1}{N}\sum_{i=1}^{N} \varphi_{F,\text{eff}}(Z, \widehat{\beta}) = \frac{1}{N}\sum_{i=1}^{N} \left(Z_i - \widehat{\beta}\right) \quad \Rightarrow \quad \widehat{\beta} = \frac{1}{N}\sum_{i=1}^{N} Z_i = \overline{Z}$$

5. Find conditions under which $\widehat{\tau}$ is RAL

- Under the conditions for the CLT:

$$\sqrt{N}\{\widehat{\tau} - \beta(F)\} \to N(0, C_{\mathcal{F}}(F)) \ , \quad C_{\mathcal{F}}(F) = \text{Var}_F\left\{\varphi_{F,\text{eff}}(Z)\right\} = \text{Var}_F(Z)$$

Example: Population Mean in the Nonparametric Model

- This implies that $\overline{Z}$ is an EIF-based estimator for $\mathrm{E}_F(Z)$ in the nonparametric model, i.e.,
- $\overline{Z}$ is robust to bias in the nuisance component estimators;
  - In this example, there is no nuisance component to be estimated, so this property is less pronounced
- $\overline{Z}$ is achieves the SEB, meaning that

1. it is the most efficient RAL estimator of $\mathrm{E}_F(Z)$ in the nonparametric model; and
2. further efficiency gains require imposing stronger assumptions on the model

Example: ATE the Nonparametric Model

1. Specify a causal or statistical model $\mathcal{F}$

- For the causal model, we assume:
  - SUTVA: $Y = Y(A)$
  - Cond. Ign.: $Y(a) \perp\!\!\!\perp A|X$
  - Positivity: $\Pr(A|X) \in (0,1)$

- The corresponding observed-data model is nonparametric, except for the positivity assumption:

$$\mathcal{F} = \left\{F \,\middle|\, f(a \mid X) \in (0,1)\right\}, \quad f = \text{density of } F$$

- It is important to note that the assumption on the counterfactual data does not impose a restriction on the observed data distribution

Example: ATE the Nonparametric Model

2. Identify a functional $\tau(F)$ that characterizes the causal parameter of interest

- The ATE is $\mathrm{E}_F\{Y(1) - Y(0)\}$
- It suffices to consider $\tau(F) = \mathrm{E}_F\{Y(1)\}$
- Under the **causal** assumptions, we have established the identification result:

$$\tau(F) = \mathrm{E}_F\{Y(1)\} = \mathrm{E}_F\big\{\mathrm{E}_F(Y|A=1,X)\big\}$$

Example: ATE the Nonparametric Model

3. Derive an IF, or if possible, the EIF, for $\tau(F)$ within the model $\mathcal{F}$

- Consider $\mathcal{F}_{\text{sub}} = \{F_\theta\}$ with $F_{\theta *} = F^*$ and $\tau(F_\theta) = \mathrm{E}_\theta\{\mathrm{E}_\theta(Y|A=1,X)\}$

- Calculate the gradient:

$$
\begin{aligned}
&\frac{\partial}{\partial \theta^{\intercal}} \tau(F_\theta) \\
&= \frac{\partial}{\partial \theta^{\intercal}} \mathrm{E}_\theta\{\mathrm{E}_\theta(Y|A=a,X)\} \\
&= \frac{\partial}{\partial \theta^{\intercal}} \int \left\{ \int y f_Y(y|A=1,x;\theta)\, dy \right\} f_X(x;\theta)\, dx \\
&= \int \left\{ \int y \nabla_\theta f_Y(y|A=1,x;\theta)\, dy \right\} f_X(x;\theta)\, dx + \int \left\{ \int y f_Y(y|A=1,x;\theta)\, dy \right\} \nabla_\theta f_X(x;\theta)\, dx \\
&= \int \left\{ \int y \frac{\nabla_\theta f_Y(y|A=1,x;\theta)}{f_Y(y|A=1,x;\theta)} f_Y(y|A=1,x;\theta)\, dy \right\} f_X(x;\theta)\, dx + \int \left\{ \int y f_Y(y|A=1,x;\theta)\, dy \right\} \frac{\nabla_\theta f_X(x;\theta)}{f_X(x;\theta)} f_X(x;\theta)\, dx \\
&= \int \left\{ \int y S_Y(y|A=1,x;\theta) f_Y(y|A=1,x;\theta)\, dy \right\} f_X(x;\theta)\, dx + \int \left\{ \int y f_Y(y|A=1,x;\theta)\, dy \right\} S_X(x;\theta) f_X(x;\theta)\, dx \\
&= \mathrm{E}_\theta\big[\mathrm{E}_\theta\{Y S_Y(Y|A=1,X;\theta) \,|\, A=1,X\} + \mathrm{E}_\theta(Y|A=1,X) S_X(X;\theta)\big]
\end{aligned}
$$

Example: ATE the Nonparametric Model

- Properties of score functions:

$$\mathrm{E}[S(W|V)|V] = 0 \qquad\qquad S(W,V) = S(W|V) + S(V)$$

- Thus, by letting $O = (Y, A, X)$

$$
\begin{aligned}
&\mathrm{E}_\theta\big[\mathrm{E}_\theta\{YS_Y(Y|A=1,X;\theta)\,|\,A=1,X\} + \mathrm{E}_\theta(Y|A=1,X)S_X(X;\theta)\big] \\
&= \mathrm{E}_\theta\left[\frac{A}{\Pr_\theta(A=1|X)}\mathrm{E}_\theta\{YS_Y(Y|A,X;\theta)\,|\,A=1,X\} + \mathrm{E}_\theta(Y|A=1,X)\{S(O;\theta) - S(Y,A|X;\theta)\}\right] \\
&= \mathrm{E}_\theta\left[\frac{A}{\Pr_\theta(A=1|X)}\mathrm{E}_\theta\big[Y\{S(O;\theta) - S(A,X;\theta)\}\,|\,A=1,X\big] + \mathrm{E}_\theta(Y|A=1,X)\{S(O;\theta) - S(Y,A|X;\theta)\}\right] \\
&= \mathrm{E}_\theta\left[\frac{A}{\Pr_\theta(A=1|X)}\mathrm{E}_\theta\{YS(O;\theta)\,|\,A=1,X\} - \frac{A\mathrm{E}_\theta(Y|A=1,X)}{\Pr_\theta(A=1|X)}S(A,X;\theta) + \mathrm{E}_\theta(Y|A=1,X)S(O;\theta)\right] \\
&= \mathrm{E}_\theta\left[\frac{AYS(O;\theta)}{\Pr_\theta(A=1|X)} - \frac{A\mathrm{E}_\theta(Y|A=1,X)}{\Pr_\theta(A=1|X)}\{S(O;\theta) - S(Y|A,X;\theta)\} + \mathrm{E}_\theta(Y|A=1,X)S(O;\theta)\right] \\
&= \mathrm{E}_\theta\left[\frac{AYS(O;\theta)}{\Pr_\theta(A=1|X)} - \frac{A\mathrm{E}_\theta(Y|A=1,X)S(O;\theta)}{\Pr_\theta(A=1|X)} + \mathrm{E}_\theta(Y|A=1,X)S(O;\theta)\right] \\
&= \mathrm{E}_\theta\left[\left[\frac{A\{Y - \mathrm{E}_\theta(Y|A=1,X)\}}{\Pr_\theta(A=1|X)} + \mathrm{E}_\theta(Y|A=1,X)\right]S(O;\theta)\right] \\
&= \mathrm{E}_\theta\left[\left[\frac{A\{Y - \mathrm{E}_\theta(Y|A=1,X)\}}{\Pr_\theta(A=1|X)} + \mathrm{E}_\theta(Y|A=1,X) - \tau(F_\theta)\right]S(O;\theta)\right]
\end{aligned}
$$

Example: ATE the Nonparametric Model

- Therefore, we have

$$\frac{\partial}{\partial \theta^{\intercal}} \tau(F_\theta)\Big|_{\theta=\theta^*} = \mathrm{E}_{F^*}\left[\left[\frac{A\{Y - \mathrm{E}_{F^*}(Y|A=1,X)\}}{\Pr_{F^*}(A=1|X)} + \mathrm{E}_{F^*}(Y|A=1,X) - \tau(F^*)\right] S(O;\theta^*)\right]$$

  implying that an IF of $\tau(F) = \mathrm{E}_F\{\mathrm{E}_F(Y|A=1,X)\}$ is

$$\varphi_F(O) = \frac{A\{Y - \mathrm{E}_F(Y|A=1,X)\}}{\Pr_F(A=1|X)} + \mathrm{E}_F(Y|A=1,X) - \tau(F)$$

- Recall that $\Lambda_{\mathcal{F}}(F) = \mathcal{L}_2^0(F)$, i.e., all mean zero and sq. integrable fts

- Therefore, $\varphi_F(O) \in \mathcal{L}_2^0(F)$, implying that

$$\varphi_{F,\text{eff}}(O) = \Pi_F\big[\varphi_F(O) \,\big|\, \Lambda_{\mathcal{F}}(F)\big] = \varphi_F(O) = \frac{A\{Y - \mathrm{E}_F(Y|A=1,X)\}}{\Pr_F(A=1|X)} + \mathrm{E}_F(Y|A=1,X) - \tau(F)$$

Example: ATE the Nonparametric Model

4. Construct $\widehat{\tau}$, an estimator for $\tau(F)$ based on the (E)IF

- We find an estimator of $\beta(F)$ by solving an moment equation:

$$0 = \frac{1}{N}\sum_{i=1}^{N} \varphi_{F,\text{eff}}(O_i, \widehat{\tau}) = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{A\{Y - \widehat{\mathrm{E}}(Y|A=1,X)\}}{\widehat{\Pr}(A=1|X)} + \widehat{\mathrm{E}}(Y|A=1,X) - \widehat{\tau}\right]$$

- Note that $\mathrm{E}(Y|A=1,X)$ and $\Pr(A=1|X)$ need to be estimated as well

- This implies that an EIF-based estimator for $\mathrm{E}_F\{\mathrm{E}_F(Y|A=1,X)\}$ in the nonparametric model is

$$\widehat{\tau} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{A\{Y - \widehat{\mathrm{E}}(Y|A=1,X)\}}{\widehat{\Pr}(A=1|X)} + \widehat{\mathrm{E}}(Y|A=1,X)\right]$$

Example: ATE the Nonparametric Model

5. Find conditions under which $\widehat{\tau}$ is RAL

- Under the "certain" conditions, we have

$$\sqrt{N}\{\widehat{\tau} - \tau(F)\} \to N(0, C_{\mathcal{F}}(F)) \ , \quad C_{\mathcal{F}}(F) = \mathrm{Var}_F\left\{\varphi_{F,\mathsf{eff}}(O)\right\}$$

- These conditions will be discussed further in the next lecture

- This implies that an EIF-based estimator for $\mathrm{E}_F\{\mathrm{E}_F(Y|A=1,X)\}$ in the nonparametric model is

$$\widehat{\tau} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{A\{Y - \widehat{\mathrm{E}}(Y|A=1,X)\}}{\widehat{\Pr}(A=1|X)} + \widehat{\mathrm{E}}(Y|A=1,X)\right]$$

- $\widehat{\tau}$ is robust to bias in the nuisance component estimators;

  We have visited this property aka double robustness property

- $\widehat{\tau}$ is achieves the SEB, meaning that

1. it is the most efficient RAL estimator of $\mathrm{E}_F\{\mathrm{E}_F(Y|A=1,X)\}$ in the nonparametric model; and
2. further efficiency gains require imposing stronger assumptions on the model

References

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